

Publishing and Sharing in Tableau

1- What is the key difference between publishing a dashboard on Tableau Public versus Tableau Server/Cloud in terms of accessibility and data security?

Answer :

The key difference is that dashboards published on *Tableau Public* are fully open to anyone on the internet, while dashboards on *Tableau Server/Cloud* are private, controlled, and secured for authorized users only. Tableau Public sacrifices data security for maximum accessibility, whereas Tableau Server/Cloud prioritizes governance, compliance, and restricted access.

Tableau Public:

Dashboards are publicly accessible to anyone with the link. No authentication required — anyone can view, download, or reuse the visualizations. Best suited for portfolio sharing, public research, or educational content.

Tableau Server/Cloud:

Dashboards are restricted to authorized users within an organization. Access is managed via user permissions, authentication, and group roles. Ideal for business intelligence, internal reporting, and compliance-sensitive data.

Data Security:

Tableau Public:

Data is uploaded to Tableau's public servers. No encryption or privacy controls — once published, data is exposed.

Risk: Sensitive or proprietary data should never be published here.

Tableau Server:

Hosted on-premises or in a private cloud. Organizations have full control over security protocols, compliance, and integration with identity providers. Suitable for industries with strict regulatory requirements (finance, healthcare, government).

Tableau Cloud:

Fully managed SaaS by Tableau. Provides robust encryption, GDPR compliance, SOC/ISO certifications, and regional data hosting. Tableau employees cannot access customer data unless explicitly permitted.

2- When publishing a Tableau dashboard, what happens to a live connection compared to an extract? How does this affect data refresh and performance?

Answer :

When we publish a Tableau dashboard, a *live connection* keeps querying the source database in real time, while an *extract* stores a snapshot of the data in Tableau's optimized engine. Live connections ensure up-to-date data but can slow performance, whereas extracts improve speed and interactivity but require scheduled refreshes to stay current.

Live Connection:

Real-time queries Every filter, sort, or interaction triggers a query directly against the database.

Data freshness: Always current, reflecting the latest changes in the source system.

Performance impact: Dependent on database speed and network latency. Complex dashboards can feel sluggish.

Best use case: Mission-critical dashboards like fraud detection, trading monitors, or incident response where data must be live.

Extract:

Snapshot stored locally Data is pulled into Tableau's Hyper engine, optimized for fast queries.

Data freshness: Static until refreshed. Refreshes can be scheduled (full or incremental).

Performance impact: Faster, smoother interactivity since queries run against Tableau's in-memory engine.

Best use case: Business dashboards viewed daily/weekly, trend analysis, or when performance is more important than second-by-second updates.

Trade-offs & Risks:

Live connections can overwhelm databases not designed for high-frequency queries, leading to delays or failures.

Extracts may become outdated if refresh schedules aren't properly managed, risking decisions on stale data.

Large extracts can also take time to refresh depending on dataset size and complexity.

3- In Tableau Server or Cloud (conceptually), which feature allows administrators to control which users can view, interact with, or edit a published dashboard?

Answer :

In Tableau Server or Tableau Cloud, the feature that allows administrators to control which users can view, interact with, or edit a published dashboard is permissions.

How Permissions Work:

User-level permissions:

Admins can grant or restrict access for individual users. Controls include viewing, interacting (filtering, drilling down), or editing/publishing dashboards.

Group-level permissions:

Permissions can be applied to groups (e.g., Finance team, Marketing team). Ensures consistent access control across departments.

Content-level permissions:

Applied directly to dashboards, workbooks, or data sources. Allows fine-grained control over who can download data, modify visualizations, or share content.

Example Scenarios:

A Finance dashboard might be restricted so only the Finance group can view and edit, while other departments can only view.

A Public KPI dashboard could allow all employees to view but restrict editing to admins.

4- If we want users to view a dashboard but not download the underlying data, which permission setting would we adjust in Tableau Server?

Answer :

To prevent users from downloading the underlying data while still allowing them to view a dashboard in Tableau Server, we need to adjust the Download permission capability — specifically disable “Download Full Data” and “Download Summary Data” for the workbook or view. This ensures viewers can interact with the dashboard but cannot export or retrieve the raw dataset.

Key Permission Settings in Tableau Server:

Download Full Data:

Controls whether users can export the entire dataset behind a visualization. Disable this to prevent raw data extraction.

Download Summary Data:

Allows exporting aggregated or summarized data. Disable this if we want to restrict all forms of data download.

View:

Must remain enabled so users can still see and interact with the dashboard.

Web Edit and Save:

Optional- disabling these prevents users from modifying or saving copies of the workbook.

Example Permission Matrix:

Capability	Allow	Deny	Effect
View			Users can see dashboards
Download Full Data			Prevents raw data export
Download Summary Data			Prevents aggregated data export
Web Edit			Stops editing in browser
Save			Stops saving new versions

Risks & Considerations:

Sensitive Data Exposure: Even if downloads are disabled, users may still screenshot or manually copy values.

Row-Level Security (RLS): Combine permissions with row-level security to ensure users only see data relevant to them.

Project-Level Locking: If permissions are locked at the project level, changes must be made by the administrator.

Balance Usability vs. Security: Some business users may need summary exports — consider enabling only “Download Summary Data” if appropriate.

5- Name two methods by which a Tableau dashboard can be embedded into a web page or website.

Answer :

Two common methods to embed a Tableau dashboard into a web page or website are:

Embedding Methods:

Embed Code (iFrame):

Tableau provides an Embed Code option when we publish a dashboard.

This generates an HTML <iframe> snippet that can be placed directly into a web page. Simple and quick, but limited in customization.

JavaScript API:

Tableau offers a powerful JavaScript API that allows developers to embed dashboards with advanced interactivity. We can control filters, parameters, events, and even integrate Tableau visuals with other web applications. Best for dynamic, customized user experiences.

6- List two best practices we should follow when preparing a dashboard for public sharing to ensure data security and privacy.

Answer :

When preparing a Tableau dashboard for public sharing, we need to balance accessibility with strict data protection. Two best practices are:

Best Practices for Data Security & Privacy:

Remove Sensitive Data:

Strip out confidential fields (e.g., customer names, IDs, financial account numbers) before publishing.

Use aggregated measures instead of raw transactional data.

Apply row-level security or filters to ensure only non-sensitive, anonymized data is exposed.

Publish Extracts Instead of Live Connections

Replace live database connections with curated extracts containing only the necessary fields.

This prevents external viewers from accessing the underlying source system.

Extracts can be refreshed securely on the server without exposing raw data.

Additional Considerations:

Disable Download Permissions- turn off “Download Full Data” and “Download Summary Data” so public users cannot export datasets.

Use Tableau Public Wisely- never publish proprietary or compliance-sensitive dashboards here.

Apply Data Masking- anonymize or obfuscate identifiers to protect privacy.

7- After publishing a dashboard, users report a “Data not found” error. What is the most common cause related to data source settings?

Answer :

The most common cause of a “Data not found” error after publishing a dashboard is related to data source credentials and gateway settings.

When we build a dashboard in tools like Poour BI, our local or desktop environment may have access to the data source (Excel file, SQL Server, API, etc.). But once published to the service, the dashboard relies on the configured data source settings in the cloud.

Common Cause: Data Source Credentials:

Invalid credentials: The account used to connect to the source is missing, expired, or not authorized in the service.

Gateway not configured: For on-premises sources (like SQL Server or local files), a gateway must be installed and mapped. Without it, the service cannot reach the data.

Different environment paths: File paths or local connections (like C:\Users\...) work on our machine but are inaccessible from the cloud.

Privacy level mismatch: Conflicting privacy settings can block queries from combining data sources.

Quick Fix Checklist:

Re-enter or update data source credentials in the service. Ensure the on-premises data gateway is installed and mapped correctly. Replace local file paths with cloud-accessible sources (OneDrive, SharePoint, SQL Azure). Verify scheduled refresh settings and confirm the service can access the source.

8- When embedding a Tableau dashboard into a secure internal portal, what two additional security measures should be implemented to ensure only authorized users can access the embedded content?

Answer :

When embedding a Tableau dashboard into a secure internal portal, we need to go beyond the default embedding setup to ensure that only authorized users can view the content. The two most important additional security measures are:

Key Security Measures:

Single Sign-On (SSO) Integration:

Configure Tableau Server/Cloud to use our organization's identity provider (e.g., Azure AD, Okta, LDAP). This ensures that users accessing the embedded dashboard are authenticated through the same secure login system as our internal portal. Prevents unauthorized external access by tying dashboard visibility to corporate credentials.

Row-Level Security (RLS):

Implement user-based filters or security tables in Tableau to restrict what data each authenticated user can see. Even if a user gains access to the dashboard, they only see the subset of data they are authorized for (e.g., region-specific, department-

specific). Protects sensitive data by enforcing fine-grained access control.

Why These Two Are Critical:

SSO ensures that only authenticated employees can even reach the embedded dashboard.

RLS ensures that once inside, users only see the data they are permitted to view.

Together, they provide end-to-end protection: authentication at the portal level and authorization at the data level.